

Radiolarian biozonation for the upper Quaternary in the Japan Sea

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Abstract

Rapid and drastic changes in radiolarian assemblages during the last 33 kyr were revealed in three sediment cores from the eastern and northeastern Japan Sea. Such changes occurred synchronously over the Sea according to the correlation by TL layers and AMS¹⁴C dating. On the basis of these bio-events, four radiolarian biozones were established: *Ceratospyris borealis* Interval Biozone (33 to 15¹⁴C kyr BP), *Stylochlamydidium venustum* Interval Biozone (15 to 12¹⁴C kyr BP), *Cycladophora davisiana* Interval Biozone (12 to 10.5¹⁴C kyr BP), and *Larcopyle butschlii* Interval Biozone (10.5¹⁴C kyr BP to Present). These radiolarian biozones are useful for correlating the upper Quaternary sediments in the Japan Sea, where carbonate fossils are poorly preserved below the calcium carbonate compensation depth (CCD).

Key words : Japan Sea, Radiolaria, upper Quaternary, Stratigraphy

Introduction

The Japan Sea is a semi enclosed marginal sea of the North Pacific and has a mean depth of 1350 m and a maximum depth of 3700 m (Fig. 1). This sea is connected with other marginal seas and with the Pacific Ocean by four shallow straits less than 130 m deep. The modern calcium carbonate compensation depth (CCD) of the Japan Sea is at around 2000 m water depth, which is remarkably shallower than what is typical in modern other oceans (Ujiié and Ichikura, 1973). Consequently, it is difficult to determine the depositional age of the bottom sediments of the Japan Sea below the CCD by ¹⁴C dating owing to poor preservation of foraminiferal fossils. Although the ¹⁴C age can also be measured on organic carbon in the sediments, terrigenous organic matter contains carbon older than the true age of the deposits (Ikehara, 2000). Therefore, the age model for the late Quaternary in the Japan Sea has built using the lithostratigraphy and tephrochronology (e.g., Tada et al., 1999).

In areas covered with carbonate-poor sediments such as the North Pacific Ocean and the Southern Ocean, the relative abundance curve of the radiolarian species *Cycladophora davisiana* has been used to establish high-resolution stratigraphy in the late Quaternary (e.g., Morley et al., 1982; Brathauer et al., 2001). Because Radiolaria comprise a group of marine planktonic Protista with robust skeletons composed of opal, they are well preserved in bottom sediments below the CCD. Therefore, they should also be useful as a stratigraphic

tool for the carbonate-poor sediments of the Japan Sea.

Studies of late Quaternary radiolarians in the Japan Sea have been limited to the southern (Sakai, 1984; Morley et al., 1986) and eastern (Itaki, 2001a) parts of the Sea. In the southern part, evident changes in the radiolarian fauna relating to environmental variations have been found (Sakai, 1984; Morley et al., 1986). Itaki (2001a) proposed radiolarian biozones for the last 30 kyr for a core from the eastern part of the Sea. However, the correlation of these sequences with other areas of the Sea has not yet been confirmed. In this report, stratigraphic changes in the radiolarian assemblage are examined for three sediment cores from the eastern and northeastern parts of the Japan Sea in order to establish the biozonation. The radiolarian biozones may be anticipated developing a more detailed age model of the Japan Sea sediments, in addition to the lithostratigraphy and tephrochronology.

Core lithology and age controls

Three piston and gravity cores obtained from the eastern and northeastern Japan Sea were examined in this study (Fig. 1). The sediment cores are mainly composed of silt or clay, and are characterized by distinct alternations of dark and light layers (Fig. 2). The dark layers are mostly laminated but sometimes contain burrows, and called TL layers (Tanaka, 1984). The TL layers of the late Quaternary are well-known ubiquitous layers that extend over the basins of the Japan Sea (Tada et al., 1992), and they have been numbered

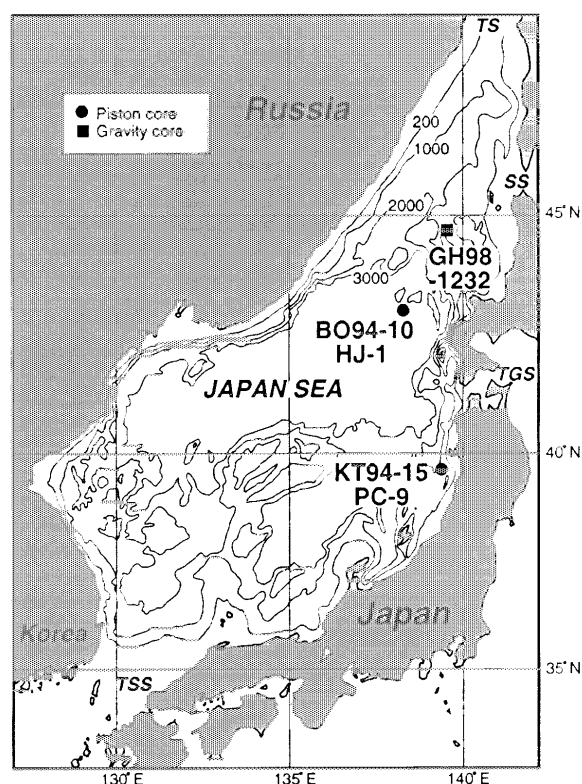


Fig. 1. Map of the Japan Sea showing locations of cores. The four straits are the Tatarskiy Strait (TS), Soya Strait (SS), Tsugaru Strait (TGS), and Tsushima Strait (TSS).

from TL-1 to TL-21 in ascending order above the Aso-4 ash layer (Tada et al., 1999). The ^{14}C ages of the younger TL layers have been previously determined using planktonic foraminifera: TL-1, $10-10.5$ kyr BP; and the top of TL-2, 15 kyr BP (Oba et al., 1995; Ikehara et al., 1996). These two horizons are therefore useful as key stratigraphic markers. In this study, the depositional ages and the correlation of horizons among the cores were estimated mainly on the basis of the TL layers and acceleration mass spectrometry (AMS) ^{14}C dating. Conventional ^{14}C ages, *i.e.*, not converted to calendar ages, are used in this study because the magnitude of the reservoir effect is unknown in this sea. Constant linear sedimentation rates between the age control points such as the TL layers and ^{14}C ages were assumed to calculate the ages of each biostratigraphic datum (Fig. 3).

Piston core KT 94-15-PC 9 (520 cm in length) was obtained from the eastern margin of the Japan Sea (lat. $39^{\circ}34'$ N, long. $139^{\circ}24'$ E) at a water depth of 807 m by the R/V *Tansei-Maru* of the University of Tokyo. It is composed of homogeneous, burrowed or thinly laminated sediments with three volcanic ashes at 63, 94, and 363 cm. Eight TL layers were observed in this core. TL-1 and the top of TL-2 are at 119-130 cm and 202 cm depth, respectively. The uppermost part of TL-2 contains burrows. Five AMS ^{14}C ages on planktonic foraminifera have been previously determined from

this core (Crusius et al., 1999).

Gravity core GH 98-1232 (437 cm in length) was obtained from the western slope (water depth, 838 m) of the Musashi Bank in the northeastern Japan Sea (lat. $44^{\circ}48'$ N, long. $139^{\circ}42'$ E) by the R/V *Hakurei-Maru* of the Geological Survey of Japan. It is composed of homogeneous, burrowed, or thinly laminated sediments. Six TL layers and two dark layers with burrows were recognized in this core. TL-1 and the top of TL-2 layers were recognized at 194-201 cm and 268 cm depth, respectively. The upper part of TL-2 contains burrows. Nineteen AMS ^{14}C ages on planktonic foraminifera have been obtained from this core (Table 1).

Piston core BO 94-10 HJ-1 (189 cm in length) was obtained from the northern part of the Japan Basin in the northeastern Japan Sea (lat. $43^{\circ}02'$ N, long. $138^{\circ}28'$ E; water depth 3613 m) by the R/V *Bosei-Maru* of Tokai University. It is composed of homogeneous or thinly laminated sediments. Two laminated layers at 104-112 cm and below 164 cm correspond to TL-1 and TL-2. The upper boundary of TL-2 is not bioturbated, and it is sharp. Two volcanic ash layers are intercalated at 16-28 cm and 178-180 cm. No ^{14}C dates have been obtained from this core owing to poor carbonate preservation.

Sample preparation procedures

Sediment samples (0.2-2.5 cm thick) were collected at intervals between 0.5 and 25 cm. In all, 135 samples (46 from PC-9, 48 from HJ-1, and 41 from GH 98-1232) were analyzed to estimate the radiolarian abundance (Table 2). Because a reduction in radiolarian shell counts in a sediment sample can be caused by oven- or air-drying processes, freeze-drying (for cores HJ-1 and GH 98-1232) or non-drying techniques (for core PC-9) were used for the quantitative analysis, as described by Itaki and Hasegawa (2000). Wet or freeze dried materials were passed through a sieve with $63\text{ }\mu\text{m}$ openings. Carbonate (which was recognized in the washed residue) was removed from carbonate-rich samples by using HCl solution. The remaining particles ($>63\text{ }\mu\text{m}$ acid insoluble residue) were pipetted onto a slide glass. After drying, they were mounted with Entellan-new.

Radiolarian species were identified and counted under an optical microscope at 100 X or 200 X magnifications. Radiolarian specimens on each slide were counted until more than 100 individuals for core GH 98-1232 and more than 300 individuals for cores PC-9 and HJ-1 had been identified (Table 2). When radiolarian concentration was too low, all specimens in two or three slides of the sample were observed. Relative and absolute abundances of radiolarians were calculated from these data.

Stratigraphic changes in radiolarian fauna

Fig. 2 shows stratigraphic changes in radiolarian fauna and

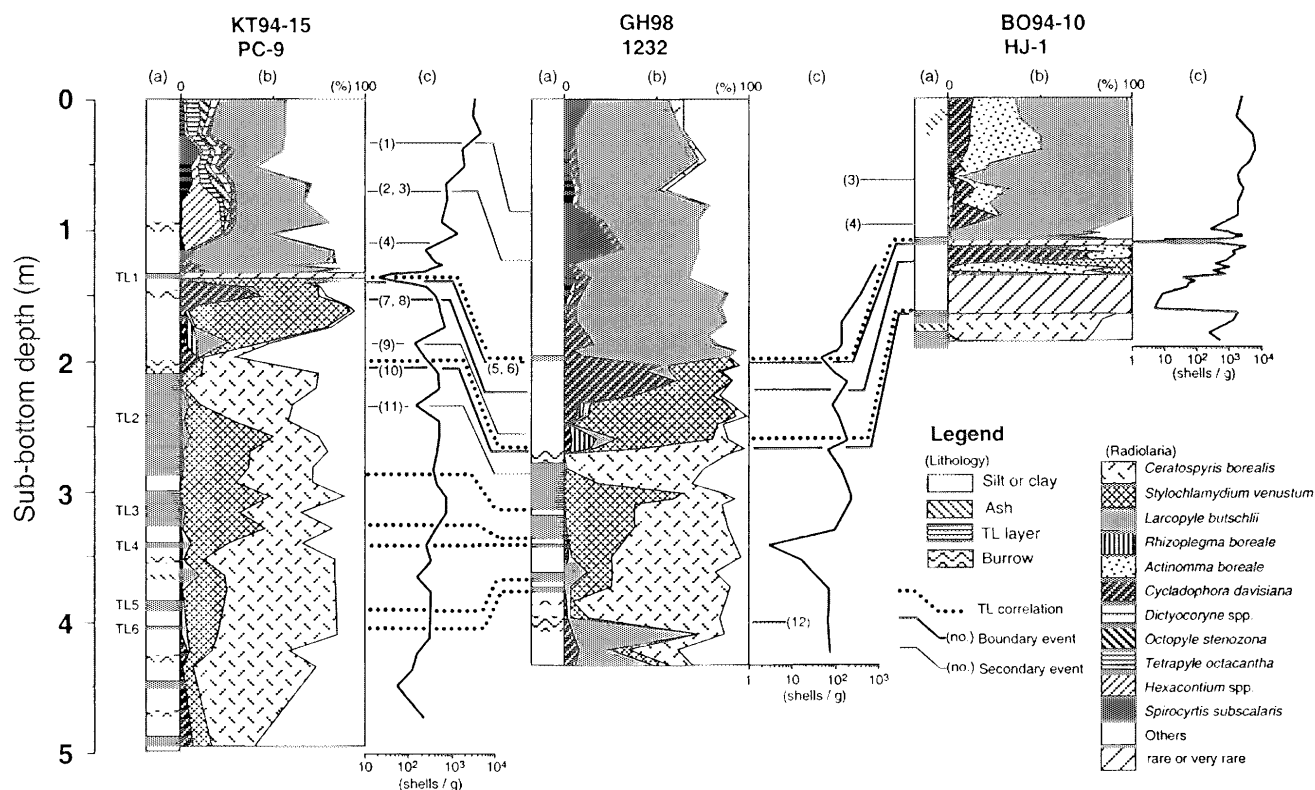


Fig.2. Correlation of radiolarian events and TL layers among cores PC-9, 1232, and HJ-1. (a) Core lithology, (b) stratigraphic changes of radiolarian fauna, and (c) radiolarian shell concentration.

shell concentrations for the three examined cores. The radiolarian assemblages were dominated by *Larcopyle butschlii* Dreyer (Figs. 4 a, 4 b), *Cycladophora davisiana* Ehrenberg (Figs. 4-c, 4 d), *Stylochlamyidium venustum* (Bailey) (Fig. 4-n), and *Ceratospirys borealis* Bailey (Figs. 4-l, 4 m). The *Actinomma boreale* Cleve group (Fig. 4 i) was also dominant in core HJ-1. The assemblages tended to be dominated by one or two species, and they varied abruptly and dramatically with age. The detailed stratigraphic changes in the assemblages are described below for each core.

1. KT94 15 PC 9

The radiolarian assemblages were dominated by *C. borealis*, which accounted for 30%–70% of the fauna in the lower and middle parts of this core (below 210 cm). In addition to *C. borealis*, *S. venustum* was also dominant, accounting for 25%–50%, between 329 and 247 cm. *Ceratospirys borealis* abruptly decreased at 195–210 cm, corresponding to the top of TL-2. A small peak in *L. butschlii* and *Rhizoplegma boreale* (Cleve) (Fig. 4-o) was recognized at 183 cm. *Stylochlamyidium venustum* abruptly increased at 171 cm, and accounted for more than 70% of the fauna until 154 cm. The abundance of *C. davisiana* was 30%–40% between 146 and 139 cm. The radiolarian shell abundance decreased remarkably from 137 to 129 cm, corresponding to TL-1. *Larcopyle butschlii* accounted for 20%–75% of the

radiolarian assemblage above 127 cm. Warm-water species such as *Tetrapyle octacantha* Muller (Fig. 4 g), *Octopyle stenozona* Haeckel (Fig. 4-h), and *Dictyocoryne* spp. occurred continuously above 74 cm. *Hexacanthium* spp. (20%) and *Spirocyrtilis subscalaris* Nigrini (6%–11%) (Figs. 4-e, 4-f) increased from 98 to 74 cm and from 62 to 39 cm, respectively.

2. GH 98 1232

Larcopyle butschlii (mostly immature form) dominated throughout the lowest part of this core, below 411 cm. The radiolarian assemblages were dominated by *C. borealis*, accounting for 20%–90% of the fauna, between 398 and 272 cm. In addition to *C. borealis*, *S. venustum* also dominated, accounting for 40%–60%, between 329 and 302 cm, but it was rare or common below 340 cm. *Ceratospirys borealis* abruptly decreased at 261–268 cm, corresponding to the top of TL-2. *Larcopyle butschlii* and *R. boreale* were transiently common at 261 cm. *Stylochlamyidium venustum* dominated from 250 to 234 cm, where they accounted for 70%–80%. A characteristic peak of *C. davisiana* (30%–40%) was recognized between 226 and 203 cm. The radiolarian abundance was rather low at 197 cm. *Larcopyle butschlii* rapidly increased at 197 cm, corresponding to TL-1, and accounted for 40%–90% of the fauna to the core top. *Spirocyrtilis subscalaris* occurred commonly (15%–30%) from

Table.1. AMS¹⁴C ages determined on planktonic foraminifera from Core GH 98-1232 from the northeastern Japan Sea. Measured radiocarbon ages, $\delta^{13}\text{C}$ values, and conventional radiocarbon ages (not corrected reservoir age) are shown.

Depth (cm)	¹⁴ C age (measured)	$\delta^{13}\text{C}$ (‰)	¹⁴ C age (conventional)	Accession No.	Material
27-29	1250 ± 60	-0.5	1650 ± 60	Beta- 134178	mixed species
42-44	1840 ± 50	0.1	2250 ± 50	Beta- 134179	mixed species
94-96	3670 ± 60	-0.5	4070 ± 60	Beta- 134180	mixed species
106-108	3860 ± 60	-0.3	4270 ± 60	Beta- 134181	mixed species
127-129	5470 ± 70	-0.3	5880 ± 70	Beta- 134182	mixed species
176-178	8590 ± 110	-0.7	8990 ± 110	Beta- 134183	mixed species
191-193	9770 ± 90	-0.8	10170 ± 90	Beta- 134184	mixed species
193-195	9760 ± 90	-1.1	10150 ± 90	Beta- 134185	<i>Globigerina bulloides</i>
196-198	9980 ± 120	-0.8	10380 ± 120	Beta- 134186	<i>Globigerina bulloides</i>
213-215	10550 ± 90	-1.0	10950 ± 90	Beta- 134187	mixed species
220-222	10500 ± 100	-1.1	10890 ± 100	Beta- 134188	mixed species
290-292	17250 ± 130	-0.5	17650 ± 130	Beta- 134189	<i>Globigerina bulloides</i>
295-297	17900 ± 180	-0.1	18310 ± 180	Beta- 134190	<i>Globigerina bulloides</i>
300-302	18450 ± 130	-1.3	18840 ± 130	Beta- 134191	<i>Globigerina bulloides</i>
311-313	20200 ± 140	0.2	20620 ± 140	Beta- 134192	<i>Globigerina bulloides</i>
319-321	20900 ± 350	-1.1	21290 ± 350	Beta- 134193	<i>Globigerina bulloides</i>
328-330	22300 ± 170	-0.8	22700 ± 170	Beta- 134194	mixed species
331-333	22700 ± 270	-0.3	23110 ± 270	Beta- 134195	<i>Globigerina bulloides</i>
404-406	32700 ± 440	-0.3	33110 ± 440	Beta- 134196	mixed species

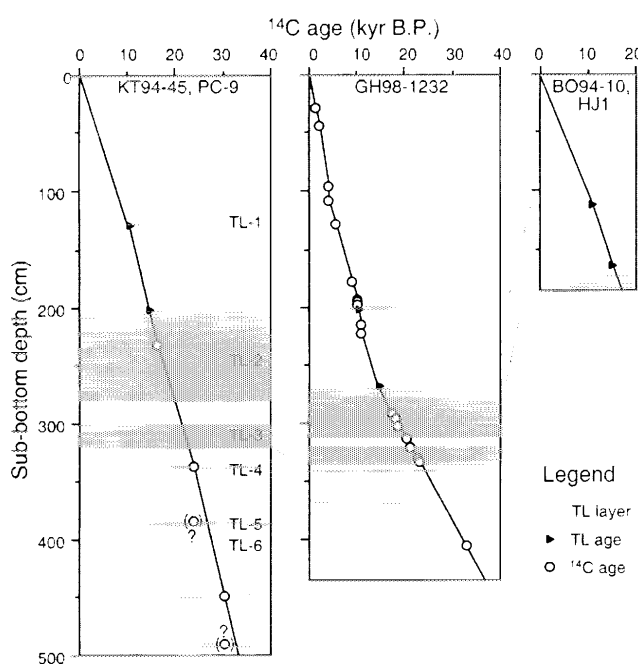


Fig.3. Sedimentation rates based on AMS¹⁴C dating (not corrected reservoir age), bottom of the TL-1 and top of the TL-2 for cores KT 94-15-PC 9, GH 98-1232 and BO 94-10-HJ 1. The ¹⁴C ages for core PC 9 have been used data from Crusius et al. (1999). Two ¹⁴C ages of 383 cm and 489 cm for core PC 9 were omitted from the age model due to their unreasonable sedimentation rates. Shaded zones indicate the thin laminated dark layer (TL layer) intervals.

115 to 93 cm.

3. BO 94-10-HJ 1

Ceratospirys borealis reached 70%-90% of the fauna in the lowest part of this core, and abruptly decreased at 161-164 cm, corresponding to the top of TL 2. Between 161 and 138 cm, radiolarian shells were very rare, although *L. butschlii* occurred commonly at 145-140 cm (Table 2). From 135 to 125 cm, *S. venustum*, *C. davisiana*, and the *A. boreale* group occurred commonly to abundantly, and the faunal composition varied distinctly: *C. davisiana* (70%) at 135 cm, the *A. boreale* group (70%-90%) between 133 and 130 cm, and *S. venustum* (50%-60%) from 128 to 125 cm. *Cycladophora davisiana* reached 50%-90% between 123 and 112 cm, and the *A. boreale* group was also abundant in this interval. Radiolarian shells were very rare or absent between 111 and 109 cm, corresponding to TL 1. *Larcopyle butschlii* accounted for 50%-90% of the fauna in the upper part of the core above 108 cm. *Heracontium* spp. was present at low levels (2%-3%) from 90 to 70 cm. The *A. boreale* group occurred abundantly above 80 cm, except for between 65 and 60 cm.

Radiolarian biozones

We were unable to recognize the stratigraphic datum planes used in Neogene biostratigraphic studies of the Japan Sea, as shown by Motoyama (1996), presumably because the present cores are too young. Then, we distinguished biozones by characteristic changes of the dominant radiolarian species, and defined the upper and lower limits of the zones. The upper and lower limits are same to the horizons of rapid increase (RI) or

Table.2. The occurrence of selected radiolarian species from cores KT 94-15, PC 9, BO 94-10, HJ-1, and GH 98-1232.**KT94-15, PC-9**

Depth in core (cm)	1	15	27	39	51	62	74	86	98	110	120	122	124	127	129	132	134	137	139	146	154	158
Radiolarian taxa																						
<i>Actinomma boreale</i>				1									1						1			
<i>Ceratospyris borealis</i>	1	2	1	3		3					5	6	5	7	3	2		3	2	7	2	5
<i>Cycladophora davisiana</i>		2	6	21	12	6	6	8	10	3	17	54	21	21	1	3	1	12	50	69	29	5
<i>Dictyocoryne</i> spp.	15	9	13	12	19	12	1	5	3													
<i>Hexacantium</i> spp.	4	6	8	4	11	13	50	43	92	1		4										
<i>Larcopyle butschlii</i>	75	150	184	81	74	109	97	99	96	62	245	272	211	68	5	3	1	2	8	5	3	4
<i>Octopyle stenozona</i>	15	14	9	8	8	15	11		2				1									
<i>Rhizoplegma boreale</i>				1														1	2	5	2	
<i>Spirocyrtis subscalaris</i>	2	7	8	35	19	17	1	1		2		1										
<i>Stylochlamydidium venustum</i>						4	2			2	13	2	1	1	5	19	13	47	55	54	312	186
<i>Tetrapyle octacantha</i>	13	29	19	15	13	18	6	3	1		1	1		2					1			
Total counted #	208	385	424	326	327	278	247	198	416	84	332	482	278	125	39	39	18	84	152	171	378	212
Rads concentration (shell / g)	1707	1841	2310	1154	1482	681	665	487	901	198	798	1021	732	336	98	106	43	254	432	347	906	422

BO94-10, HJ-1

Depth in core (cm)	1	16	29	40	50	55	60	65	70	80	90	101	103	103.5	104.3	105.3	106.3	107.3	108.2	108.8	109.1	109.5
Radiolarian taxa																						
<i>Actinomma boreale</i>	74	59	306	135	75	52	19	18	120	80	7											1
<i>Ceratospyris borealis</i>	2	1			1	3									1	2						
<i>Cycladophora davisiana</i>	42	26	87	35	21	32	3	36	61	19	80	6	1	6	3	13	7	5	3	4		2
<i>Dictyocoryne</i> spp.	1		3		2		1	2	1	2												
<i>Hexacantium</i> spp.								4	22	12	9						1					
<i>Larcopyle butschlii</i>	200	129	385	164	349	333	321	327	427	394	244	33	189	274	250	612	179	474	369	39		
<i>Octopyle stenozona</i>	2	1	1	1	1	4			4	3												
<i>Rhizoplegma boreale</i>																						
<i>Spirocyrtis subscalaris</i>	1					2			3													
<i>Stylochlamydidium venustum</i>	1																					
<i>Tetrapyle octacantha</i>			3	1	5	1	4	4	2										1			
Total counted #	331	221	795	340	463	436	353	397	654	517	342	41	227	303	304	795	393	510	384	44	1	2
Rads concentration (shell / g)	2489	1700	5230	6054	4132	2295	1810	1909	2616	1814	1754	241	841	1027	1328	2291	2530	1393	1289	121	10	23

GH98-1232

Depth in core (cm)	1	24	47	70	81	93	115	126	137	148	159	170	181	188	193	197	204	209	215	227	234	243
Radiolarian taxa																						
<i>Actinomma boreale</i>																						
<i>Ceratospyris borealis</i>	10	2	4	4	2						2	1		3			2	2		4	3	7
<i>Cycladophora davisiana</i>			5		1	6	6	4	10	5	5	8	13	24	6	7	41	50	70	46	9	9
<i>Dictyocoryne</i> spp.	1																					
<i>Hexacantium</i> spp.				3	3					4	1	1		1								
<i>Larcopyle butschlii</i>	50	60	66	45	80	70	47	80	65	100	85	100	70	100	100	50	3	14	2		5	4
<i>Octopyle stenozona</i>	1																					
<i>Rhizoplegma boreale</i>																					4	4
<i>Spirocyrtis subscalaris</i>	16	6	1	6	5	22	32	2	8	3	1		1				1					
<i>Stylochlamydidium venustum</i>										1		1		1	1		29	30	35	50	90	100
<i>Tetrapyle octacantha</i>			1		1	1				1												
Total counted #	120	105	100	105	116	137	118	123	124	129	111	126	108	154	115	64	80	112	117	119	123	126
Rads concentration (shell / g)								871		386		136		139	105	45		98	187		84	121

rapid decrease (RD) in the species abundance, which are defined in Motoyama (1996).

Radiolarian events, such as the RI, RD, and maximum occurrence of marker species, are listed in Table 3 together with their estimated ^{14}C ages. The events are numbered as shown in Fig. 2 and Table 3. Four radiolarian biozones were recognized in the late Quaternary Japan Sea sediments of the last 33 ^{14}C kyr BP. Their boundary and secondary radiolarian events are basically the same as those recognized by Itaki (2001a) in core PC-9 with minor revisions from bottom to top as follows.

1. *Ceratospyris borealis* Interval Biozone

Definition : <Bottom> RI of *Ceratospyris borealis* (event no. 12) ; <Top> RD of *Ceratospyris borealis* (event no. 10).

Age : 33 to 15 ^{14}C kyr BP.

Secondary radiolarian event : RD of *S. venustum* (17.5 ^{14}C kyr BP) (event no. 11).

Faunal character : The assemblage is characterized by the dominance of *C. borealis*. *Stylochlamydidium venustum* is also dominant in the middle part of this biozone but tends to decrease below TL-4. This species is absent or very rare in the upper part of TL-2.

Remarks : The top of the *C. borealis* Zone corresponds to the top of TL-2 (Fig. 2 ; Table 3). The bottom of this biozone coincides with the RD of *L. butschlii* (immature form) in core GH 98-1232. Cores PC-9 and HJ-1 did not reach it. Bottom of core HJ-1 corresponds to the upper part of this biozone above the RD of *S. venustum*, which is subzone-a of the *Ceratospyris borealis* assemblage zone of Itaki (2001a).

2. *Stylochlamydidium venustum* Interval Biozone

Definition : <Bottom> RD of *Ceratospyris borealis*

171	183	195	210	222	234	247	259	271	283	295	305	317	329	341	354	365	378	412	424	437	449	474	498
11	2	29	127	310	23	97	48	137	86	140	142	152	74	76	156	54	138	131	56	50			7
12	1	14				1	1	2				3		1	1	1	1	1	5	1			2
		1					2		2					1									
7	8	3	2	35	2	24	7	13	3	12		33	2	9	3	8	2	6	2				
17	3	1	1																				
2																							
311	9	5	20	9	3	86	86	118	88	103	170	138	91	37	23	13	55	33	15	4			3
437	46	171	201	435	46	325	193	324	250	338	352	462	203	182	217	88	230	201	120	67			28
795	113	395	365	1142		93	673	333	721	422	677	524	874	319	351	380	161	368	391	230	129	58	

1103	1113	1118	1123	113	114	116	118	120	123	125	128	130	133	135	138	140	143	145	150	155	161	164	170	180	186
	1	8	11	72	116	166	52	81	76	32	92	262	132	41	5	5		1	4	1					
		1	6	17	1	3	2		4	1	2						3				2	821	463	109	190
	3	109	238	407	317	160	314	244	278	88	17	1	3	173	14	3	1	1		1	1	3	1		
												9		2		16	38	26		1					
							1										1								
				1				6	168	102	26	43	20	1		1								17	
0	4	118	265	536	438	334	369	326	364	290	213	298	178	236	20	24	44	28	4	3	3	882	554	139	278
0	17	388	866	1581	3172	1893	1183	1516		775	336	1024	289	789	33	88	52	47	9	6	5	1764	1291	232	518

250	261	268	272	284	295	302	306	311	329	340	352	363	375	386	398	411	423	436
3	9	46	100	100	60	24	50	38	68		30	80	70	67	80	18	30	16
3	5	3		2			1		2		1			1	2	6	9	8
1	11	2									1	18	4	2	2	85	19	57
	14	9																
85	55	22		1	22	70	40	70	50		9	20	24	1	11		12	12
107	106	84	113	136	118	109	106	117	135		43	144	109	81	113	126	113	133
	193	68		140		230	227			95		17		69	71	65		76

(event no. 10) : <Top> RD of *Stylochlamydidium venustum* (event no. 8).

Age : 15 to 12 ¹⁴C kyr BP.

Secondary radiolarian events : Small peaks of *L. butschlii* and *R. boreale* (14 ¹⁴C kyr BP) (event no. 9).

Faunal character : The assemblage of this biozone is occupied mostly by *S. venustum* in cores PC 9 and GH 98-1232, and is characteristically accompanied by the *A. boreale* group or *C. davisiana* in core HJ-1. In the lower part of this biozone, *L. butschlii* and *R. boreale* commonly occur in cores PC 9 and GH 98-1232. Radiolarian abundance decreases in the lower part of this biozone in core HJ-1.

Remarks : The bottom of this biozone corresponds to the top of TL-2. The *S. venustum* Zone corresponds to subzone-b of the *stylochlamydidium venustum* assemblage zone of Itaki (2001a).

3. *Cycladophora davisiana* Interval Biozone

Definition : <Bottom> RD of *Stylochlamydidium venustum* (event no. 8) : <Top> RD of *Cycladophora davisiana* (event no. 6).

Age : 12 to 10, 5 ¹⁴C kyr BP.

Secondary radiolarian event : RD of *C. davisiana* (12 ¹⁴C BP) (event no. 7).

Faunal character : This biozone is characterized by the abundant occurrence of *C. davisiana*. The assemblage is accompanied by *S. venustum* in cores PC 9 and GH 98-1232 and by the *A. boreale* group in core HJ-1.

Remarks : This biozone is recognized only within a narrow interval of sediments just below TL-1 (Fig. 2 : Table 3). The bottom of this biozone coincides with the RD of *C. davisiana* (event no. 7). Radiolarian abundance tends to decrease just above this biozone. The top of this biozone corresponds to the

Table.3. Horizons (cm) of radiolarian bio-events and age controls in cores KT 94-15, PC-9, BO 94-10, HJ-1, and GH 98-1232. AMS¹⁴C ages in Core PC-9 have been referred to Crusius et al. (1999).

Radiolarian Zones	Radiolarian events and age control horizons	KT94-15 PC-9	GH98 1232	BO94-10 HJ-1	AMS C-14 (kyr BP)
<i>L. butschlii</i> Zone	AMS ¹⁴ C		28		1.7
	AMS ¹⁴ C		43		2.3
	(1) RD <i>S. subscalaris</i>	27-39	81-93		ca. 2-4
	AMS ¹⁴ C		95		4
	AMS ¹⁴ C		107		4.3
	(2) RI <i>S. Subscalaris</i>	63-74	115-126		ca. 5.5
	(3) RD <i>Hexacontium</i> spp.	63-74		65-70	ca. 6
	AMS ¹⁴ C		128		5.9
	AMS ¹⁴ C		177		8.9
	(4) RI <i>Hexacontium</i> spp.	98-110		90-101	ca. 8.5
	AMS ¹⁴ C		192		10.2
	AMS ¹⁴ C		194		10.2
<i>C. davisiana</i> Zone	AMS ¹⁴ C		197		10.4
	(5) RI <i>L. butschlii</i>	127-129	197-204	108-109	ca. 10.5
	Bottom TL-1	130	201	112	10.5
	(6) RD <i>C. davisiana</i>	137-139	197-204	111-112	ca. 10.5
<i>S. venustum</i> Zone	AMS ¹⁴ C		214		11
	AMS ¹⁴ C		221		10.9
	(7) RI <i>C. davisiana</i>	146-154	227-234	125-128	ca. 12
<i>C. borealis</i> Zone	(8) RD <i>S. venustum</i>	146-154	227-234	123-125	ca. 12
	(9) Peak <i>L. butschlii</i> / <i>R. boreale</i>	183	261		ca. 14
<i>C. borealis</i> Zone	(10) RD <i>C. borealis</i>	195-210	261-268	161-164	ca. 15
	Top TL-2	202	268	164	15
	AMS ¹⁴ C	231			16.2
	AMS ¹⁴ C		291		17.7
	(11) RD <i>S. venustum</i>	234-247	284-295		ca. 17.5
	AMS ¹⁴ C		296		18.3
	AMS ¹⁴ C		301		18.8
	AMS ¹⁴ C		312		20.6
	AMS ¹⁴ C		320		21.3
	AMS ¹⁴ C		329		22.7
	AMS ¹⁴ C		332		23.1
	AMS ¹⁴ C	337			24
	AMS ¹⁴ C	383			23.9
	AMS ¹⁴ C	447			30.4
	AMS ¹⁴ C	489			30.6
	(12) RI <i>C. borealis</i>		398-411		ca. 33
	AMS ¹⁴ C		405		33.1

bottom of the interval in which radiolarians are very rare in cores PC-9 and HJ-1, and to the RI of *L. butschlii* (event no. 5) in core GH 98-1232. The *C. davisiana* Zone corresponds to subzone-a of the *Stylochlamydidium venustum* assemblage zone of Itaki (2001 a).

1. *Larcopyle butschlii* Interval Biozone

Definition : <Bottom> RD of *Cycladophora davisiana* (event no. 6).

Age : 10.5 ¹⁴C kyr BP to present.

Secondary radiolarian events : RI of *L. butschlii* (10.5 ¹⁴C kyr BP) (event no. 5), RI of *Hexacontium* spp. (8.5 ¹⁴C kyr BP) (event no. 4), RD of *Hexacontium* spp. (6 ¹⁴C kyr BP) (event no. 3), RI of *S. subscalaris* (5.5 ¹⁴C kyr BP) (event no. 2), and RD of *S. subscalaris* (2-4 ¹⁴C kyr BP) (event no. 1).

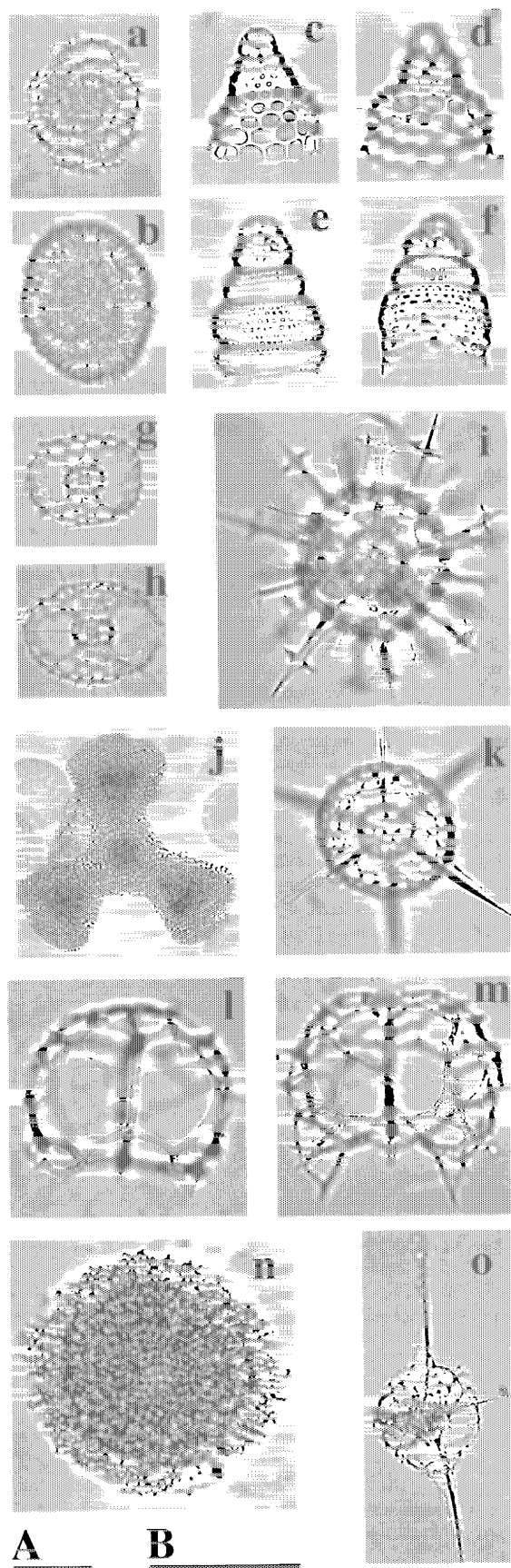
Faunal character : The assemblage is characterized by the dominance of *L. butschlii*. Warm-water species such as *T. octacantha*, *O. stenozona*, and *Dictyocoryne* spp. occur

commonly in core PC-9 but sporadically in cores GH 98-1232 and HJ-1. The *A. boreale* group is abundant in core HJ-1, but absent or sporadic in cores PC-9 and GH 98-1232. *Cycladophora davisiana* is common to abundant in core HJ-1, while it is rare in cores PC-9 and GH 98-1232 except in the lowest part of this biozone. The abundant occurrence of *Hexacontium* spp. is observed in the lower middle part of this biozone in core PC-9. *Spirocorytis subscalaris* maximums are recognized in the middle part of this biozone in cores PC-9 and GH 98-1232.

Remarks : Radiolarian abundance decreases in the lowest part of the *L. butschlii* Zone, corresponding to TL-1 in cores PC-9 and HJ-1 (Fig. 2 : Table 3). The bottom of this biozone corresponds to the RI of *L. butschlii* (event no. 5) in core GH 98-1232.

Discussion and conclusions

Our results indicate that rapid radiolarian faunal changes



occurred simultaneously in the northeastern (cores GH 98-1232 and HJ 1) and eastern Japan Sea (core PC 9). Four biozones were established based on these radiolarian events. The radiolarian biozones have also been correlated between core PC 9 (eastern Japan Sea) and core KH 79-3, C 3, obtained from the Oki Ridge in the southern Japan Sea (Itaki, 2001a). Recently, similar radiolarian faunal changes were reported in a core obtained from the Wakasa Basin of the southern Japan Sea (Hoshiba et al., 2001). These results indicate that these radiolarian biozones can be widely correlated in the Sea and that they are thus useful stratigraphic tool for upper Quaternary Japan Sea sediments, most of which are carbonate-free sediments below the CCD.

Such the rapid changes in the radiolarian assemblage as observed in the Japan Sea have not been reported from the other ocean basins. This means that the application of the radiolarian biozones proposed in this report is limited within the Japan Sea. Morley et al. (1986) correlated the maximum occurrence of *C. davisiana* at 302 cm depth in core RC 12-379 from the southern part of the Japan Sea to the last glacial maximum (LGM) by comparing it with the observed occurrence pattern over the northwest Pacific (Morley et al., 1982). Our result, however, suggests that the most recent peak of *C. davisiana* in the Japan Sea should be correlated with the latest Pleistocene (12-10, 5 $^{\circ}\text{C}$ kyr BP : *C. davisiana* Zone), and not with the LGM. During the LGM, the deep-water anoxic condition was caused by strong stratification with input of fresh water to the sea-surface in the Japan Sea (Oba et al., 1991, 1995). Although *C. davisiana* adapts to the highly oxygenated deep-water in the modern Japan Sea (Itaki, in press), it might not be able to survive in oxygen poor water during the LGM.

The *A. boreale* group is abundant and shows stratigraphic changes characteristically in core HJ-1 from 3613 m depth. However, it is almost absent or sporadic in cores PC 9 and GH 98-1232 from about 800 m depth. This occurrence pattern is

Fig.4. Photomicrographs of late Quaternary radiolarians from the Japan Sea :

a and b, *Larcopyle butschlii* Dreyer (a, PC 9, 2-11 ; b, PC-9, 2-11) ; c and d, *Cycladophora davisiana* Ehrenberg (c, PC 9, 2-16 ; d, PC 9, 3-32) ; e and f, *Spirocorythis subscalaris* Nigrini (e, PC 9, 2-16 ; f, PC 9, 2-16) ; g, *Tetrapyle octacantha* Müller, (PC 9, 2-16). h, *Octopyle stenozona* Haeckel, (PC 9, 2-16) ; i, *Actinomma boreale* Cleve, (PC -9, 2-16) ; j, *Dictyocoryne truncatum* (Ehrenberg), (PC -9, 2-16) ; k, *Hexacontium pachydermum* Jørgensen, (PC -9, 3-2) ; l and m, *Ceratospyrus borealis* Bailey (l, PC 9, 5-13 ; m, PC 9, 4-5) ; n, *Stylochlamydidium venustum* (Bailey), (PC 9, 4-11) ; o, *Rhizoplegma boreale* (Cleve), (PC 9, 3-37). Scale bars A (for a, b, c, d, and j) and B (c-f, i and k-o) show 100 μm .

related to the discrete depth distribution of the *A. boreale* group, which lives at depths greater than 2000 m in the present Japan Sea (Itaki, in press). It is expected, therefore, that the abundance curve of the *A. boreale* group can be used as a stratigraphic tool only for cores from greater depths.

Because *C. davisiana* is also a deep-water species that lives mainly below 1000 m depth in the modern Japan Sea (Itaki, in press), the abundance of this species is smaller in cores PC-9 and GH 98–1232, which are from shallower depths, than in core IJJ-1, which is from a deeper depth. Moreover, *C. davisiana* occurred only sporadically during the last 24.5 kyr BP in a core from 283 m water depth off the Tango Peninsula in the southern Japan Sea, probably because the water depth was too shallow (Itaki, 2001b). These facts suggest that the *C. davisiana* Zone, which is characterized by abundant *C. davisiana*, cannot be distinguished in cores from shallow water.

Taxonomic remarks

Actinomma boreale Cleve group (Fig. 4-i) : According to Cortese and Björklund (1998), *Actinomma boreale* Cleve and *Actinomma leptoderma leptoderma* (Jørgensen) differ in the number of spines and the presence of the 4th cortical shell. In the present study, *A. l. leptoderma* has been included in the *Actinomma boreale* Cleve group as a juvenile form of *A. boreale* owing to similarities between *A. l. leptoderma* and *A. boreale* and the existence of transitional forms between the two species. Most of the *A. boreale* group is composed of *A. boreale*, although several percent of this species group are evidently properly classified as *A. l. leptoderma*.

Ceratospyris borealis Bailey (Figs. 4-l, 4-m) : Most specimens of *C. borealis* in this study have a somewhat thinner trellis-type skeleton compared with typical forms found in the northern Pacific, the Bering Sea, and the Sea of Okhotsk (Nigrini, 1970; Ling et al., 1971; Ling, 1974).

Larcopyle butschlii Dreyer (Figs. 4-a, 4-b) : This species varies from 60 to 230 μm in length. There are two morphological types of *L. butschlii* : with complete shell (adult form; Fig. 4-b) and incomplete shell (immature form; Fig. 4-a). The immature form is similar to the inner structure of the adult form. In general, the radial spines of the adult form are fewer than those of the immature form. In core GH 98–1232, the immature form dominated throughout the lowest part (below 411 cm).

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- **in Japanese

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日本海東部および北東部より得られた3本の海底コアから過去3万3千年間の放散虫群集の突然かつ急激な変化を明らかにした. このような変化は日本海のほぼ全域で同時に起こったことを示しており, そのバイオ・イベントをもとに4つの放散虫化石帯を設定した. すなわち, *Coratospyris borealis* 間隔帯 (33~15 kyr BP), *Stylochlamyidium venustum* 間隔帯 (15~12 kyr BP), *Cycladophora davisiana* 間隔帯 (12~10.5 kyr BP), *Larcopyle butschlii* 間隔帯 (10.5 kyr~現在) である. これらの放散虫化石帯区分は, 炭酸塩補償深度 (CCD) 以深の炭酸塩化石の産出が少ない日本海の後期第四紀堆積物の対比に有効である.